Lecture 27

Improper integrals and the Gamma function

Uses.

Summary 2: Every nonempty real set bounded above has a supremum. The Archimedean property supplies arbitrarily large integers.
Summary 5: One-sided and infinite-endpoint limits obey the algebraic and order laws when their finite limits exist.
Summary 6: The standard limits include $\sin x/x \to 1$ and $(e^x - 1)/x \to 1$ as $x \to 0$.
Summary 8: Elementary derivatives, including real powers on positive arguments, obey the chain and product rules.
Summary 11: Finite Taylor expansion of the exponential has a Lagrange remainder; for $t > 0$ its terms and remainder are positive.
Summary 25: Finite Riemann integrals obey order, additivity, linearity and the absolute-value bound.
Summary 26: Continuous functions are integrable on closed finite intervals. The fundamental theorem evaluates them by primitives and differentiates their accumulation integrals. Definite substitution and parts hold with their stated continuity hypotheses.

Objectives. *An integral over an infinite interval, or past an unbounded endpoint, is a limit of finite integrals. We determine when that limit exists before computing it. Power models and comparison then justify an integral definition of the Gamma function and its recurrence.*

By the end of this lecture you will be able to:

- Define convergence at each improper endpoint separately.
- Prove convergence or divergence using power models and comparison.
- Justify the Gamma integral and derive its recurrence by finite-cutoff parts.
- Differentiate integral-defined functions and distinguish absolute from conditional convergence.

27.1 Limits of finite integrals

Until now both endpoints were finite and the integrand was bounded. We keep the finite integral and take a limit of its endpoints. Throughout this lecture, every integrand is Riemann integrable on each closed finite subinterval that avoids its specified exceptional endpoints. All the examples below are continuous on those subintervals.

Definition 27.1. For a function $f : [a, \infty) \to \mathbb{R}$, the *improper integral* $\int_a^\infty f(x)\,\mathrm{d}x$ *converges* if there is $L \in \mathbb{R}$ such that

$$\lim_{R\to\infty} \int_a^R f(x)\,\mathrm{d}x = L.$$

Its value is then L. Otherwise it *diverges*. For a function $f : (a, b] \to \mathbb{R}$, with $a < b$ finite, convergence at the excluded endpoint a means that the following limit is finite:

$$\int_a^b f(x)\,\mathrm{d}x := \lim_{c\downarrow a} \int_c^b f(x)\,\mathrm{d}x.$$

Convergence at an excluded upper endpoint is defined by $c \uparrow b$. At a lower infinite endpoint, define $\int_{-\infty}^b f(x)\,\mathrm{d}x = \lim_{R\to-\infty} \int_R^b f(x)\,\mathrm{d}x$, again requiring a finite limit.

A limit equal to $+\infty$ or $-\infty$ is divergence, even when its sign is known. For $\int_{-\infty}^\infty f$, split at a finite regular point and require both integrals to converge. At an interior exceptional point, split there and require the two one-sided integrals separately. If there are several exceptional points, every resulting piece must converge.

The regular splitting point does not change the value or convergence. For example, for $a < d < R$, finite additivity gives $\int_a^R f = \int_a^d f + \int_d^R f$. The first term is fixed and finite. The limit law for addition proves the claim for the tail; the same argument applies to the other endpoint types.

Figure 27.1 shows two finite integrals before their limits are taken. The shaded regions always have ordinary finite endpoints.

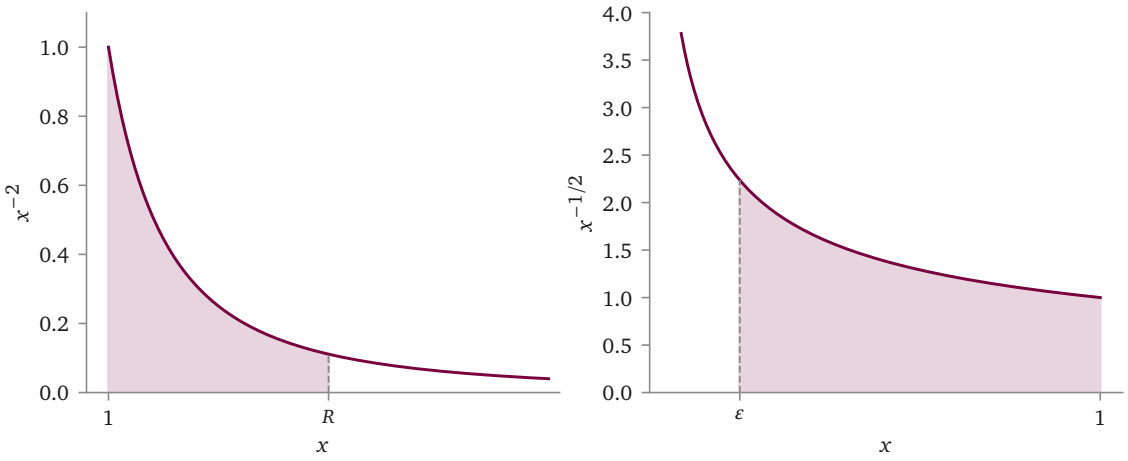


Figure 27.1: A tail cutoff R for x^{-2}, and a lower cutoff ε for $x^{-1/2}$. The respective limits move R to infinity and ε down to zero.

Example 27.2 (Two divergent pieces cannot cancel)**.** Determine whether $\int_{-1}^1 \mathrm{d}x/x$ exists as an improper integral. The exceptional point is zero. For $0 < \varepsilon < 1$, the two finite pieces are

$$\int_{-1}^{-\varepsilon} \frac{\mathrm{d}x}{x} = \ln \varepsilon, \qquad \int_\varepsilon^1 \frac{\mathrm{d}x}{x} = -\ln \varepsilon.$$

The first tends to $-\infty$ and the second to $+\infty$, so the improper integral diverges. Their sum for matching cutoffs is zero, but this coupled limit does not satisfy the definition requiring each piece to converge.

27.2 The power models

Fix $p \in \mathbb{R}$. On positive arguments, the power and logarithm derivatives give the following finite-cutoff formulas, with $R > 1$ and $0 < \varepsilon < 1$:

$$\int_1^R x^{-p}\,\mathrm{d}x = \begin{cases} (R^{1-p}-1)/(1-p), & p \neq 1, \\ \ln R, & p = 1, \end{cases} \qquad \int_\varepsilon^1 x^{-p}\,\mathrm{d}x = \begin{cases} (1-\varepsilon^{1-p})/(1-p), & p \neq 1, \\ -\ln \varepsilon, & p = 1. \end{cases}$$

The fundamental theorem applies on every such finite interval. If $p > 1$, then $R^{1-p} \to 0$; if $p < 1$, then $R^{1-p} \to \infty$. For the lower endpoint, $\varepsilon^{1-p} \to 0$ precisely when $p < 1$. At $p = 1$ the logarithms tend to $+\infty$ in both expressions. Thus the complete convergence test is

Integral	Converges exactly when	Value when convergent
$\int_1^\infty x^{-p}\,\mathrm{d}x$	$p > 1$	$1/(p-1)$
$\int_0^1 x^{-p}\,\mathrm{d}x$	$p < 1$	$1/(1-p)$

All remaining cases diverge to $+\infty$. The direction of the inequality reverses between the two endpoints: a power can be harmless near zero and too large at infinity. For example, $x^{-1/2}$ is integrable near zero, while x^{-2} is integrable at infinity.

27.3 Nonnegative integrands and comparison

Theorem 27.3 (Bounded cutoff criterion)**.** *Let $f \geq 0$ on $[a, \infty)$. Its improper integral converges if and only if its finite-cutoff values $\int_a^R f$ are bounded above. The corresponding criterion holds at a finite excluded endpoint.*

Proof. Put $A(R) = \int_a^R f$. If $R_2 > R_1 \geq a$, finite additivity and order give $A(R_2) - A(R_1) = \int_{R_1}^{R_2} f \geq 0$. Suppose the cutoff values are bounded, and let S be their supremum. For $\eta > 0$, the definition of supremum supplies R_0 with $S - \eta < A(R_0) \leq S$. For every $R \geq R_0$, monotonicity gives $S - \eta < A(R) \leq S$. This is $A(R) \to S$. Conversely, convergence bounds the tail of the cutoff values, and monotonicity bounds all earlier values by any one later value.

At an excluded lower endpoint, put $B(c) = \int_c^b f$. As c decreases towards a, this value increases. If the values are bounded, take their supremum S and choose c_0 with $B(c_0) > S - \eta$. For $a < c \leq c_0$, the same inequalities give $B(c) \to S$. The converse and the upper-endpoint case follow in the same way. ∎

For signed integrands, boundedness alone is insufficient. Indeed, $\int_0^R \sin x\,\mathrm{d}x = 1 - \cos R$ is bounded, but its values at $R = 2n\pi$ and $R = (2n+1)\pi$ are 0 and 2. It has no limit.

Theorem 27.4 (Comparison and limit comparison)**.** *Suppose $0 \leq f \leq g$ sufficiently near one improper endpoint. If the integral of g converges there, so does that of f. If the integral of f diverges there, so does that of g. If instead $f, g > 0$ near that endpoint and $f/g \to L$ for $L \in (0, \infty)$, then their integrals either both converge or both diverge.*

Proof. Discard a fixed finite interval to reach the neighbourhood on which the inequality holds. Every cutoff integral of f is between zero and the corresponding integral of g. A finite limit for the nonnegative g provides an upper bound, so Theorem 27.3 gives convergence of

f. The divergence implication is the contrapositive. If $f/g \to L > 0$, then sufficiently near the endpoint

$$\frac{L}{2}g \le f \le \frac{3L}{2}g.$$

The two comparison directions prove equivalence. Multiplication by a positive constant does not affect convergence, by the finite integral's linearity and the limit law for constant multiples. ∎

A zero limit does not give equivalence. For instance, $f(x) = x^{-2}$ and $g(x) = x^{-1}$ have $f/g \to 0$ at infinity, but only the integral of f converges.

Example 27.5 (Recognising the endpoint power)**.** Test the convergence of the following two integrals:

$$\int_1^\infty \frac{3x^2+1}{x^4+4}\,\mathrm{d}x, \qquad \int_0^1 \frac{e^x-1}{x^{3/2}}\,\mathrm{d}x.$$

For the first, divide the integrand by x^{-2}. The ratio is $x^2(3x^2+1)/(x^4+4) \to 3$. The power model at infinity has $p = 2 > 1$, so the integral converges. For the second, divide by $x^{-1/2}$. The ratio is $(e^x-1)/x \to 1$ by Lecture 6. The lower-endpoint power has $p = 1/2 < 1$, so this integral also converges. Both comparisons use positive integrands in the endpoint neighbourhoods.

27.4 Absolute convergence and evaluation

Definition 27.6. An improper integral of f *converges absolutely* if the integral of $|f|$ converges at every improper endpoint. It *converges conditionally* if the integral of f converges but that of $|f|$ does not.

Lecture 25 proves that $|f|$ is integrable on every finite interval on which f is Riemann integrable.

Theorem 27.7 (Absolute convergence implies convergence)**.** *If the improper integral of $|f|$ converges, that of f converges and $|\int f| \le \int |f|$, with the same endpoint ranges on both integrals.*

Proof. Define $f_+ = (|f|+f)/2$ and $f_- = (|f|-f)/2$. Both are nonnegative, integrable on each finite cutoff interval by linearity, and bounded above by $|f|$. Comparison therefore gives finite improper integrals of both. The identity $f = f_+ - f_-$ and the limit law for differences give convergence of f. Pass to the limit in the finite absolute-value inequality to obtain the stated bound. For several improper pieces, apply this argument to each piece and then use the triangle inequality. ∎

The integral $\int_1^\infty \sin x/x\,\mathrm{d}x$ converges conditionally. Sheet 27 proves this by finite integration by parts and the lower bound $|\sin x| \ge \sin^2 x$. Its tail also satisfies $|\int_R^\infty \sin x/x\,\mathrm{d}x| \le 2/R$ for $R \ge 1$.

Example 27.8 (Separate limits after substitution)**.** Evaluate the integral

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x}(1+x)}.$$

Near zero, its ratio to $x^{-1/2}$ tends to 1. At infinity, its ratio to $x^{-3/2}$ tends to 1. Both power tests converge, so both improper pieces exist. For $0 < \delta < R$, use $x = t^2$ on $[\sqrt{\delta}, \sqrt{R}]$. Its derivative $2t$ and the positive branch give

$$\int_\delta^R \frac{\mathrm{d}x}{\sqrt{x}(1+x)} = 2\int_{\sqrt{\delta}}^{\sqrt{R}} \frac{\mathrm{d}t}{1+t^2} = 2\arctan\sqrt{R} - 2\arctan\sqrt{\delta}.$$

Splitting at 1 and taking each endpoint limit separately gives the value $2(\pi/2) - 0 = \pi$. The transformation was applied only to finite intervals; the endpoint limits were justified afterwards.

The same practice applies to parts: first use the definite rule on finite cutoffs, then check the limits of every resulting boundary and integral term.

Convergence can also give a quantitative error bound for a finite calculation. For the preceding integrand $f(x) = 1/(\sqrt{x}(1+x))$, positivity gives $f(x) \le x^{-1/2}$ and $f(x) \le x^{-3/2}$ for every $x > 0$. Use the first bound near zero and the second at infinity. If $0 < \delta \le 1 \le R$, additivity and the two convergent power integrals give

$$0 \le \pi - \int_\delta^R f(x)\,\mathrm{d}x = \int_0^\delta f(x)\,\mathrm{d}x + \int_R^\infty f(x)\,\mathrm{d}x \le 2\sqrt{\delta} + \frac{2}{\sqrt{R}}.$$

These are ordinary separate tails, obtained by passing to their finite-cutoff bounds. Thus an error tolerance $\eta > 0$ is guaranteed by choosing $\delta < (\eta/4)^2$ and $R > (4/\eta)^2$, as well as $\delta \le 1 \le R$. The bound is conservative but explicit: it controls what is omitted, without treating a plotted finite interval as the whole integral.

27.5 The Gamma integral

For a real parameter $s > 0$, define the *Gamma function* by

$$\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,\mathrm{d}t. \tag{27.1}$$

We must establish convergence at both endpoints before using this definition. On $0 < t \le 1$, the integrand is at most t^{s-1}. The lower-endpoint power test converges because $s - 1 > -1$. For the tail, choose an integer $N > s + 1$, possible by the Archimedean property. The finite Taylor formula of Lecture 11, with positive terms and remainder, gives $e^t \ge t^N/N!$ for $t > 0$. Consequently

$$0 < t^{s-1}e^{-t} \le N!\,t^{s-1-N} \qquad (t \ge 1).$$

The exponent is less than -2, so comparison proves convergence at infinity. For $s \le 0$, the same integral representation diverges at zero: $e^{-t} \ge e^{-1}$ on $(0, 1]$ and t^{s-1} has exponent at most -1. Thus $s > 0$ is the exact domain of this integral definition.

Theorem 27.9 (Gamma recurrence). *For every $s > 0$, one has $\Gamma(s+1) = s\Gamma(s)$. Moreover $\Gamma(1) = 1$, so $\Gamma(n+1) = n!$ for every integer $n \ge 0$, where $0! = 1$.*

Proof. For $0 < \varepsilon < R$, finite integration by parts gives

$$\int_\varepsilon^R t^s e^{-t}\,\mathrm{d}t = [-t^s e^{-t}]_\varepsilon^R + s\int_\varepsilon^R t^{s-1}e^{-t}\,\mathrm{d}t.$$

The lower boundary term tends to zero because $s > 0$ and $e^{-t} \to 1$. For the upper one, choose an integer $N > s+1$ as above. Then $0 \le R^s e^{-R} \le N! R^{s-N} \to 0$. Both integrals already converge by the preceding argument for their positive parameters $s+1$ and s. Taking the two endpoint limits separately proves the recurrence. Finally, $\int_0^R e^{-t}\,\mathrm{d}t = 1 - e^{-R} \to 1$ gives $\Gamma(1) = 1$. Repeated recurrence gives $\Gamma(n+1) = n!$, including $n = 0$. ∎

Example 27.10 (Integer and half-integer parameters)**.** Find $\Gamma(5)$ and express $\Gamma(7/2)$ in terms of $\Gamma(1/2)$. The recurrence gives

$$\Gamma(5) = 4 \cdot 3 \cdot 2 \cdot 1 = 24, \qquad \Gamma(7/2) = \frac{5}{2}\frac{3}{2}\frac{1}{2}\Gamma(1/2) = \frac{15}{8}\Gamma(1/2).$$

To identify the remaining constant, substitute $t = x^2$ on finite positive cutoffs. Since $\mathrm{d}t = 2x\,\mathrm{d}x$ and $t^{-1/2} = 1/x$, passing to the separately convergent endpoint limits gives

$$\Gamma(1/2) = 2\int_0^\infty e^{-x^2}\,\mathrm{d}x.$$

This proves convergence of the Gaussian integral. Lecture 38 will compute its value using two-dimensional integration; that value is not assumed here.

27.6 Functions defined by integrals

An antiderivative need not have a formula built from a finite number of algebraic operations, compositions, exponentials, logarithms and trigonometric functions. Such formulas are called *elementary* here. The existence of an accumulation integral requires only continuity, by Lecture 26, regardless of whether an elementary formula is available.

Define the *error function* and the *sine integral*, respectively, for every $x \in \mathbb{R}$ by

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,\mathrm{d}t, \qquad \operatorname{Si}(x) = \int_0^x q(t)\,\mathrm{d}t, \tag{27.2}$$

where $q(0) = 1$ and $q(t) = \sin t/t$ for $t \neq 0$. The standard sine limit makes q continuous at zero, and it is continuous elsewhere by the quotient law. Both definitions therefore use proper finite integrals, with the reversal convention when $x < 0$. The fundamental theorem gives, including at $x = 0$,

$$\operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}}e^{-x^2}, \qquad \operatorname{Si}'(x) = q(x).$$

Both integrands are even, so substitution $t = -u$ in each finite integral shows that both functions are odd. Their values at zero are zero. The error function is strictly increasing because its derivative is positive. It has a finite limit at infinity by convergence of the Gaussian integral; Lecture 38 will identify that limit as 1. The sine integral has a finite limit at infinity by Sheet 27's conditional-convergence proof, after adding its proper integral over $[0, 1]$.

Figure 27.2 displays numerical curves on finite intervals.

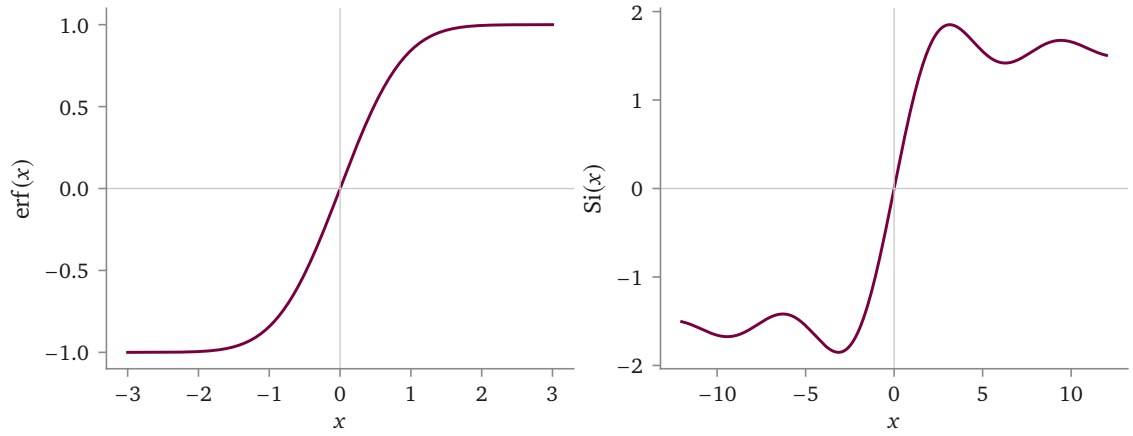


Figure 27.2: Numerical plots of the error function and sine integral on finite intervals. The sine integral's derivative changes sign; the error function's derivative remains positive.

We record as a cited result [C] that neither e^{-x^2} nor $\sin x/x$ has an elementary antiderivative on an open interval of regularity. Rosenlicht's finite-integration theorem supplies these examples; the Gaussian version follows from his exponential example by complex linear rescaling. This external result concerns elementary formulas, not the existence of the integral-defined functions. Its proof is outside this course.

Summary 27.

- An improper integral is a finite limit of finite integrals. Every exceptional endpoint must be tested separately; cancellation of divergent pieces does not count as convergence.
- The power x^{-p} is integrable at infinity exactly for $p > 1$, and at zero exactly for $p < 1$. Nonnegative cutoff values converge exactly when bounded.
- Theorems 27.4 and 27.7: positive comparison and positive finite limit comparison test convergence; absolute convergence implies convergence.
- Equation (27.1) and Theorem 27.9: the Gamma integral converges exactly for $s > 0$ and obeys $\Gamma(s+1) = s\Gamma(s)$ and $\Gamma(n+1) = n!$.
- Continuous integrands define differentiable accumulation functions, including erf and Si, even when their antiderivatives are not elementary [C].

Sheet 27 proves conditional sine-integral convergence, a limit of finite integrals and the Wallis recurrence and product. Lecture 28 uses definite integrals to compute geometric and physical quantities.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §7.8. NIST Digital Library of Mathematical Functions, definitions 7.2.1 and 6.2.9. For nonelementarity [C], see M. Rosenlicht, "Integration in Finite Terms", *American Mathematical Monthly* **79**(9) (1972), 963–972, §6, pp. 971–972.